TECH REPORT

Clinically-Informed Brugada ECG Detection: Comparing Morphology-Aware Models and Deep Learning Under Limited Data

Team BIBD

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Abstract

1. Introduction

Brugada syndrome is a cardiac condition associated with an increased risk of ventricular arrhythmias and sudden cardiac death [1]. Electrocardiography (ECG) plays a central role in its detection, particularly through characteristic patterns in the right precordial leads (V1–V3). However, these patterns can be subtle and variable, making reliable identification challenging in clinical practice.

Recent studies have explored machine learning and deep learning approaches for automated ECG-based detection. Deep learning models can learn complex representations from raw signals, but they often require large datasets and lack interpretability. In small and imbalanced datasets, such as Brugada ECG cohorts, these models are prone to overfitting and unstable performance.

An alternative approach is clinically-informed modeling through morphology-aware feature extraction. By explicitly capturing ECG characteristics such as QRS morphology, ST-segment changes, and T-wave features, these models align more closely with established clinical knowledge and may provide more robust and interpretable results under limited data conditions.

In this study, we compare morphology-aware models and deep learning models for Brugada detection using a patient-level evaluation framework. We focus on an ECG-only setting to ensure fair comparison and avoid reliance on non-signal metadata. The evaluated models include gradient boosting methods with genetic algorithm optimization and 1D convolutional neural networks using both V1–V3 and full 12-lead inputs.

Our results show that clinically-informed representations and targeted lead selection can provide competitive and more stable performance than full 12-lead deep learning models under limited data conditions. These findings highlight the importance of integrating clinical knowledge into model design and support the use of such systems as decision-support tools rather than standalone diagnostic solutions.

2. Dataset and Problem Definition

This study uses the Brugada-HUCA dataset, which consists of 12-lead ECG recordings from patients labeled as either Normal or Brugada syndrome. After excluding atypical cases, the final dataset contains 356 patients, with 287 Normal and 69 Brugada cases.

To ensure a fair and clinically realistic evaluation, a patient-level stratified split is applied. The dataset is divided into 80% training (284 patients) and 20% testing (72 patients), while preserving class distribution. The test set is strictly held out and not used during model development or threshold tuning.

The task is formulated as a binary classification problem: distinguishing Brugada from Normal ECG recordings. In this context, the objective is not only to achieve strong overall performance but also to balance sensitivity and specificity, as missing a Brugada case (false negative) carries higher clinical risk than generating a false positive.

Table 1. Dataset Summary

Variable	Value
Patients	356
Normal	287
Brugada	69
Train	284
Test	72

3. Methodology

3.1. Preprocessing

All ECG recordings were preprocessed to reduce noise and ensure consistency across patients. Signals were filtered using a bandpass filter (0.5–40 Hz) to remove baseline drift and high-frequency noise, followed by normalization. R-peaks were detected to identify individual heartbeats, and signal segments were extracted around each beat. For morphology-aware modeling, relevant ECG segments were used to compute clinically meaningful features. For deep learning models, the preprocessed signals were used directly as input.

3.2. Feature Representation

For morphology-aware models, features were extracted from ECG signals to capture clinically relevant characteristics, including QRS duration, ST-segment morphology, T-wave features, and amplitude-based descriptors. These features were computed across multiple leads, with particular emphasis on right precordial leads (V1–V3), which are known to be critical for Brugada syndrome detection.

For deep learning models, raw ECG signals were used as input. Two configurations were evaluated: (1) V1–V3 leads only, focusing on clinically relevant regions, and (2) full 12-lead ECG input to capture global signal information.

3.3. Model Families

First, classical machine learning models were used as baselines, including Logistic Regression, Random Forest, and XGBoost.

Second, morphology-aware models were implemented using gradient boosting methods, including XGBoost and LightGBM. Hyperparameter optimization was performed using a genetic algorithm to improve model performance [2].

Third, deep learning models were implemented using 1D convolutional neural networks (CNNs). Models were trained on both V1–V3 inputs and full 12-lead signals. To improve robustness, multiple training runs with different random seeds were conducted.

3.4. Evaluation Strategies

Model performance was evaluated using a patient-level split to avoid data leakage. Cross-validation was performed on the training set, while the test set was held out for final evaluation.

We report multiple evaluation metrics, including ROC-AUC, F1-score, Matthews Correlation Coefficient (MCC), sensitivity, and specificity, to capture both overall performance and clinical relevance.

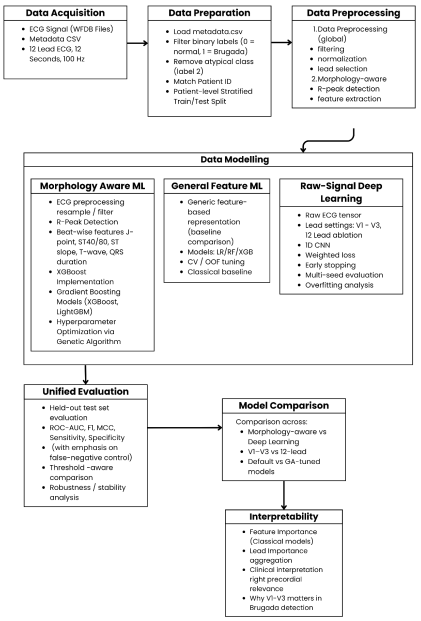

Image 1. Clinical Methodology Flowchart

4. Results

4.1. Overall Model Performance

We first compare the performance of all evaluated models under a unified test set. The results show that morphology-aware models and deep learning models achieve comparable performance, but with different trade-offs in stability and interpretability.

Among the morphology-aware approaches, gradient boosting models with genetic algorithm optimization achieved the highest overall performance. In particular, the LightGBM model with GA tuning demonstrated strong discrimination ability and balanced performance across evaluation metrics.

For deep learning models, the CNN trained on V1–V3 leads achieved competitive results, while the CNN using full 12-lead input showed unstable behavior and reduced performance. This suggests that incorporating all leads does not necessarily improve model effectiveness under limited data conditions.

4.2. Morphology Aware Models

Morphology-aware models consistently identified clinically relevant ECG features as important predictors. Models such as XGBoost and LightGBM demonstrated strong performance, with genetic algorithm optimization improving ranking-based metrics such as ROC-AUC.

However, improvements in ROC-AUC did not always translate into significant gains in threshold-based metrics such as F1-score or sensitivity. This indicates that while GA enhances model discrimination, the final classification performance remains dependent on threshold selection and class imbalance.

4.3. Deep Learning Models

Deep learning models showed different behavior depending on the input configuration. The CNN trained on V1–V3 leads achieved stable and competitive performance, highlighting the importance of clinically relevant lead selection.

In contrast, the CNN trained on full 12-lead input exhibited signs of instability, including imbalanced predictions and reduced effective performance. This suggests that the inclusion of all leads may introduce noise and increase the risk of overfitting when the dataset size is limited.

4.4. Comparative Analysis

Overall, morphology-aware models and V1–V3-based deep learning models provided the most reliable performance. Morphology-aware models offered better interpretability and consistent feature importance aligned with clinical expectations, while deep learning models benefited from direct signal representation but required careful input selection.

These results indicate that clinically-informed feature design and targeted lead selection can be as effective as, or more robust than, full 12-lead deep learning approaches in small and imbalanced ECG datasets

Table 2. Model Comparison Result

Model	ROC - AUC	F1	Sensitivity
XGBoost	0.730	0.571	0.571
LightGBM (GA)	0.862	0.714	0.714
XGBoost (GA)	0.861	0.714	0.714
CNN (V1-V3)	0.858	0.651	0.595
CNN (12-Leads)	0.840	0.326	1.000

The results indicate that morphology-aware models and V1–V3-based deep learning models provide more stable performance, while the 12-lead CNN shows inconsistent behavior.

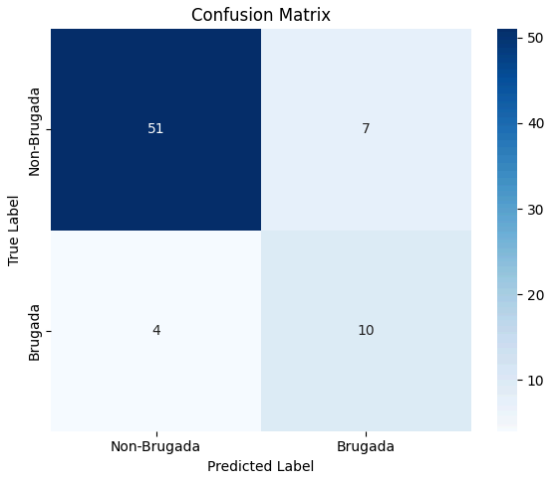

Image 2. Confusion Matrix of CNN V1-V3 Lead

To further analyze the behavior of the best-performing deep learning model, the confusion matrix of the CNN trained on V1–V3 leads is shown in Figure X. The model achieves a balanced performance, correctly identifying most Brugada cases while maintaining reasonable specificity, although several false positives and false negatives remain.

5. Interpretability Analysis

To better understand model behavior, we analyze feature importance from morphology-aware models and examine the contribution of different ECG leads.

The morphology-aware models consistently identified features related to ST-segment morphology and QRS characteristics as the most important predictors. In particular, features derived from the right precordial leads, especially V2 and V1, were among the top contributors across models. This aligns with the known clinical manifestation of Brugada syndrome, which primarily affects these leads.

Further analysis of lead-level importance confirms that V1–V3 contain the most relevant information for detection. This observation is consistent with the performance of the deep learning models, where the CNN trained on V1–V3 leads achieved more stable results compared to the full 12-lead model.

These findings suggest that focusing on clinically relevant leads and morphology-based features not only improves interpretability but also enhances robustness under limited data conditions.

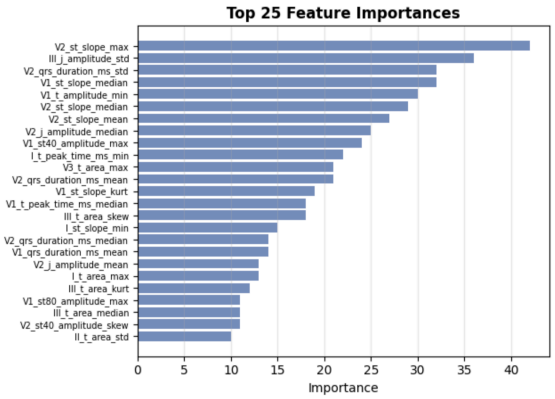

Image 3. Feature Importance of LGBM (GA)

The feature importance analysis shows that ST-segment slope-related features, particularly from lead V2, dominate the model's predictions. Features from right precordial leads (V1–V3) consistently appear among the top contributors, supporting their clinical relevance in Brugada syndrome detection.

6. Discussion

Morphology-aware models, particularly gradient boosting with genetic algorithm optimization, achieved the most consistent performance across metrics. In

contrast, deep learning models showed varying behavior depending on input configuration. The CNN trained on V1–V3 leads demonstrated stable and competitive results, while the full 12-lead model exhibited unstable behavior despite moderate ROC-AUC, suggesting increased noise and overfitting under limited data.

A gap between ranking-based and threshold-based metrics was also observed. While some models achieved high ROC-AUC, their F1-score and sensitivity did not always improve accordingly, highlighting the importance of evaluating models beyond AUC in clinical settings.

Both morphology-aware and deep learning approaches consistently emphasized the importance of right precordial leads (V1–V3), particularly lead V2. This alignment with known clinical patterns supports the use of clinically-informed modeling for robust and interpretable Brugada detection.

7. Limitations

This study has several limitations. First, the dataset size is relatively small, which may limit model generalization. Second, evaluation was conducted on a single dataset without external validation. Third, differences in feature representation and model configurations may affect the fairness of comparisons across approaches.

8. Conclusion

In this study, we compared morphology-aware models and deep learning approaches for Brugada syndrome detection using ECG data. The results show that clinically-informed feature-based models, particularly gradient boosting with genetic algorithm optimization, achieve strong and stable performance under limited data conditions.

Deep learning models demonstrated competitive results when focused on clinically relevant leads (V1–V3), but full 12-lead models showed unstable behavior, highlighting the importance of targeted input selection.

Overall, these findings emphasize that incorporating clinical knowledge into model design can improve both performance and interpretability. The proposed approaches are best positioned as decision-support tools to assist clinicians in Brugada detection.

9. References

[1] Bergonti, M., et al. (2025). The clinical significance of atrial fibrillation in non–high-risk Brugada syndrome: The BruFib study. JACC: Clinical Electrophysiology, 11(11), 2471–2480. https://doi.org/10.1016/j.jacep.2025.06.031

[2] Deng, X., Li, M., Deng, S., & Wang, L. (2022). Hybrid gene selection approach using XGBoost and multi-objective genetic algorithm for cancer classification. Medical & biological engineering & computing, 60(3), 663–681. https://doi.org/10.1007/s11517-021-02476-x